

PLACEMENT CLASS SHEET**Data Sufficiency**

Directions for the questions (1 – 40) : Select from one of the given choices.

- (A) if statement I alone sufficient to answer the question.
- (B) if statement II alone sufficient to answer the question.
- (C) if Both I & II together sufficient to answer the question.
- (D) if Neither I nor II alone is sufficient to answer the question.
- (E) if Either I or II alone is sufficient to answer the question.

1. What is the remainder if the positive integer
★ 'y' is divided by 3?

- I) 'y' is an even integer
- II) 'y' is a multiple of 6

2. What is the value of the positive two digit
★ number 'x'?

- I) The sum of its digits is 4
- II) The difference of its digits is 4

3. If bowl S contains only marbles, how many marbles are in the bowl?

- I) If $\frac{1}{4}$ of the marbles were removed, the bowl would be filled to $\frac{1}{2}$ of its capacity.

II) If 100 marbles were added to the bowl, it would be filled.

4. If John is 5 years older than Kathy, how old is John?

- I) Seven years ago, John was 3 times as old as Kathy was at that time.
- II) Kathy and John were born on the same day of the month.

5. A dress was initially listed at a price that
★ would have given the store a profit of 20 percent of the wholesale cost. What was the wholesale cost of the dress?

- I) After reducing the asking price by 10 percent, the dress sold for a net profit of 10 dollars.
- II) The dress sold for 50 dollars.

6. Car X traveled from city A to city B in 30
★ minutes. What was the average speed of car X?

- I) Half the distance of the trip was covered at 50 miles per hour, and the other half of the distance was covered at 60 miles per hour.
- II) Car X traveled at 30 miles for the full trip.

7. Steve bought some apples at a cost of \$60 each and some oranges at a cost of \$50 each.

If he paid a total of \$410, how many apples did Steve buy?

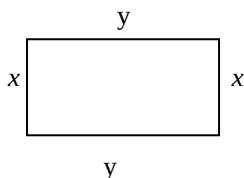
- I) The total number of apples and oranges Steve bought was 8.
- II) Steve bought seven times as many oranges as apples.

8. If $x, y \neq 0$, is $\frac{x}{y}$ an integer?

- I) x is prime
- II) y is even

★ 9. If the sides x of the rectangle below are increased by 3 units, what was the original area?

- I) The resulting area is 20.
- II) The resulting figure is a square.



10. Suppose $3p + 4q = 11$, then what is the value of q ?

- I) p is prime
- II) $q = -2p$

11. Is $x - y$ divisible by 5?

★

- I) x is divisible by 5
- II) y is divisible by 5

12. How many students are enrolled in Math?

- I) There are 72 students enrolled in History.

II) There are 40 students enrolled in both History and Math.

13. Is $x - y = 0$?

★

- I) $\frac{x}{y} > 0$
- II) $x^2 = y^2$

14. If P is a positive integer, is P even?

★

- I) P divided by 3 leaves a remainder of 1.
- II) P divided by 4 leaves a remainder of 1.

15. If $x \neq 0, -1$, then is $\frac{1}{x}$ greater than $\frac{1}{x+1}$?

- I) $x < 1$
- II) $x > 1$

16. If $x = 3y$, what is the value of $x + y$?

- I) $3y = 4x + 21$
- II) $x > y$

17. An integer greater than 1 is prime if it is divisible by itself and 1. Is n prime?

★

- I) n is between 1 and 4
- II) $n^2 - 5n + 6 = 0$

18. A piece of wood 5 feet long is cut into three smaller pieces. How long is the longest of the three pieces?

★

- I) One piece is 2 feet 7 inches long.
- II) One piece is 7 inches longer than another piece and the remaining piece is 5 inches long.

19. Two pipes, A and B, empty into a reservoir. Pipe A can fill the reservoir in 30 minutes by itself. How long will it take for pipe A & B together to fill up the reservoir?

- I) By itself, Pipe B can fill the reservoir in 20 minutes.
- II) Pipe B has a larger cross-sectional area than Pipe A.
20. If xy greater than 1? x and y are both positive numbers. ★
- I) x is less than 1
- II) y is greater than 1
21. Is Rajesh wearing a rain coat?
- I) It is raining heavily.
- II) Rajesh never goes out in rain without wearing a rain coat.
22. How many brothers does Raju have? ★
- I) Raju's two sisters are twins.
- II) Raju's uncle has only one nephew.
23. In how many days will the work done by R and S?
- I) R is twice efficient as S.
- II) S takes 5 more days than R.
24. How many members are there in committee A? ★
- I) All the members of committee B are in A.
- II) The number of members of committee B is 25.
25. What percent of all the balls in the bag were green?
- I) There were exactly 10 green balls.
- II) The ratio of red to blue balls was 8 : 13.
26. Drug addicts were arrested.
- I) Taking drugs is a crime.
- II) One who commits a crime can be arrested.
27. The use of strong hollow bricks has led to safer building.
- I) Hollow bricks by the nature of their structures are stronger than solid bricks.
- II) The stronger the building, the safer it is.
28. How many books can xyz publishing co. print in a day? ★
- I) xyz publishing co. has 15 presses.
- II) Each press can print 3000 pages everyday.
29. How much is the salary of Rakesh?
- I) Rakesh's salary is 60% of Raja's salary.
- II) Raja's salary is 70% of Vinod's salary.
30. How many books are on the book self? ★
- I) The book self is 20 metres long.
- II) The average weight of each book is $\frac{1}{4}$ kgs.
31. Is William taller than Jane?
- I) William is taller than Suresh.
- II) Suresh is not as tall as Jane.
32. Who types at a faster rate Antony or Xavier?

- I) The difference between their typing rates is 20 words per minute.
- II) Xavier types at a constant rate of 100 words per minute.
33. How many pencils does Antony have?
- I) He bought 5 boxes of pencils.
- II) He lent 2 pencils to Raja.
34. Can truck A pass safely underneath an elevated highway 12 feet above the ground?
- I) Truck B can pass safely underneath the highway.
- II) Truck B is taller than Truck A.
35. Who is the tallest of the three?
- I) x is taller than y .
- II) z is not as short as y .
36. What is today's date and month?
- ★ I) In two weeks it will be the first of a month.
- II) Two weeks ago it was the first of a month.
37. If we assume a constant reading rate, can Raja finish the book in 8 hours?
- I) Raja reads 75 pages an hour.
- II) In 3 hours Raja reads half the book.
38. Did Hindustan Lever make profit in 2000?
- I) The company made profit in 1999.
- II) The company made profit in 2001.
39. How old is Raja?
- I) 20 years from now he will be 2 years older than triple his present age.
- II) He is 3 years older than his brother.
40. How many red marbles are in a box?
- ★ I) Squaring the number of red marbles would have the same effect as tripling it.
- II) There are ten marbles in the box.

SOLUTIONS

1. Ans : B

Solution :

I) is not sufficient. For e.g., $4 \div 3$ leaves remainder 1, $6 \div 3$ leaves remainder 0.

II) is sufficient to answer the question.
Because it is a multiple of 6.

2. Ans : C

Solution :

Considering I alone, x must be 13, 22, 31 or 40.

Considering II alone, x must be 40, 51, 15, 62, 73, 26, 37, 84, 48, 95 or 59.

By combining both I & II together x is 40.

3. Ans : C

Solution :

Both I & II together are sufficient to answer the question : let n be the number of marbles in the bowl and let c be the capacity of the bowl. Then from

$$I) n - \frac{1}{4}n = \frac{1}{2}c, \text{ or } \frac{3}{2}n = c$$

Next from II) $100 + n = c$. Hence, we

$$\text{have the system : } \frac{3}{2}n = c$$

$100 + n = c$. From this we can calculate the value of n .

4. Ans : A

5. Ans : A

Solution :

From the question, we get

$$p - 0.1p = c + 10$$

$$0.9p = c + 10$$

$$p = \frac{c + 10}{0.9}$$

$$p = 1.2c \text{ given}$$

$$\frac{c + 10}{0.9} = 1.2c$$

From this we can obtain the required answer.

6. Ans : E

7. Ans : D

Solution :

$$0.60x + 0.50y = 4.10 \quad \text{Given in question}$$

$$x + y = 8 \quad \text{Given in I}$$

From this we can obtain the answer.

Also, from II alone we can obtain the answer.

$$y = 7x \quad \text{Given in II}$$

8. Ans : E

9. Ans : C

From I), the area of this figure is $y(x + 3) = 20$.

From II), if the resulting figure is square, then $y = x + 3$.

From I & II together we can obtain the answer.

10. Ans : B

11. Ans : C

Solution :

Both I & II are together sufficient to give the answer.

12. Ans : E

Solution :

From both the statements the value of Math is not clear.

13. Ans : C

Solution :

Combining both the statements we can obtain the answer.

14. Ans : B

Solution :

From statement I we cannot determine

From statement II

$p = 4 \cdot z + 1$ is odd since of is 1 more than the even expression. Hence statement II alone is sufficient.

15. Ans : D

Solution :

Using statement I

Let $x = 0.5$

$$\text{So, } \frac{1}{x} = \frac{1}{0.5} = 2; \quad \frac{1}{0.5 + 1} = \frac{1}{1.5} < 1$$

$$\text{So, } \frac{1}{x} > \frac{1}{x + 1}$$

$$x = -2 \quad \frac{1}{x} = -\frac{1}{2} = -0.5 \quad \frac{1}{-1} = -1$$

$$\frac{1}{x} > \frac{1}{x + 1}$$

Similarly we can derive using statement II. So, statement I or II alone is enough.

16. Ans : A

Solution :

It is clear from the question that statement I alone is sufficient to answer.

$$3y = 4x + 21$$

$$x = 3y$$

By solving this system, we can determine the value of both x and y and therefore the value of $x + y$. Hence statement I alone is sufficient.

17. Ans : D

Solution :

From statement I, the only integers between 1 and 4 are 2 and 3 and both are prime. Hence we can determine that 'n' is prime.

From statement II, solving the equation

$$n^2 - 5n + 6 = 0$$

$$(n - 2)(n - 3) = 0$$

$$n = 2 \text{ or } 3$$

Hence, $n = 2$ or $n = 3$ again both 2 and 3 are prime.

$\therefore$ The answer is D.

18. Ans : D

Solution :

Statement I alone is sufficient. Since 2 feet 7 inch is more than half of 5 feet, the piece that is 2 feet 7 inch long must be longer than the other two pieces put together.

Statement II alone is sufficient, since one piece is 5 inches long, the sum of the lengths of the remaining two pieces is 4 feet 7 inches. Since one piece is 7 inches longer than the other, $L + (L + 7 \text{ inches}) = 4 \text{ feet } 7 \text{ inches}$, where L is the length of the smaller of the two remaining pieces. Solving the equations yields $L + 7 \text{ inches}$ as the length of the longest pieces.

19. Ans : A

Solution :

It is clear from the question given.

20. Ans : E

Solution :

Both the statements are not sufficient to answer the question. Hence the answer is E.

21. Ans : C

Solution :

It is clear from the statement given.

22. Ans : B

Solution :

From II Raju has no brother because he is his uncle's only nephew.

23. Ans : C

Solution :

From I & II, we can find two distinct equation to find out the alone time of S and alone time of R. Hence the answer can be derived.

24. Ans : E

Solution :

From I members of B are in A. But we do not know that whether A has more members.

I) is not sufficient

II) is not sufficient

Even if we combine I & II, the data's are not sufficient.

25. Ans : E

Solution :

From I and II it is not known that red, blue and green are the only colors in the bag.

26. Ans : C

Solution :

From I & II, taking drugs is a crime and one who commits a crime can be arrested. So drug addicts were arrested.

27. Ans : C

Solution :

From I & II, Hollow bricks are stronger and the stronger the building, the safer it is.

28. Ans : E

Solution :

Using I and II one can get only the total number of pages that can be printed in xyz co. in a day but not the number of books printed as required in the question.

29. Ans : E

Solution :

Statements I and II related to Raja's salary with other salaries but none of the salary is known so we need more datas.

30. Ans : E

Solution :

Statements I and II do not give information about the width of the book. So, we need more datas.

31. Ans : E

Solution :

Data I is not enough; Data II is not enough

From I and II

William = x, Jane = y, Suresh = z

From I) $x > z$ From II) $z < y$

From this we can not conclude whether $x > y$.

32. Ans : E

Solution :

From I) alone is not sufficient to answer the question because it does not identify who types at the faster rate.

From II) alone is not sufficient.

From I & II together, Xavier's rate is 80 or 120 words per minute. We cannot derive the correct answer. We need more data.

33. Ans : E

Solution :

Without knowing the number of pencils in a box it is impossible to answer the question.

34. Ans : C

Solution :

Combining I and II, Taller truck B can pass safely. So the shorter truck A can also safely pass underneath the highway.

35. Ans : E

Solution :

From I & II x and z are taller than y . So the possibilities are x is the tallest or z is the tallest.

We cannot derive the correct answer.

We need more data.

36. Ans : C

Solution :

Combining I & II. It was clearly February 15.

37. Ans : B

Solution :

Statement I is not sufficient.

From II we can conclude that Raja takes 6 hours to finish the book.

38. Ans : E

Solution :

From year to year the profit differs. So we need more data's to derive the answer.

39. Ans : A

Solution :

From I, if x is Raja's present age then $x + 20 = 2 + 3x$

So $x = 9$

Statement I is sufficient.

Since, we do not know Raja's brother's age, II is not sufficient.

40. Ans : A

Solution :

From I $x^2 = 3x$

$$x^2 - 3x = 0$$

$$\Rightarrow x(x - 3) = 0$$

$$\Rightarrow x = 0 \text{ (or) } x = 3$$

So we can take $x = 3$

Statement I is sufficient.

ANSWER KEY

- | | |
|------|-------|
| 1. B | 26. C |
| 2. C | 27. C |
| 3. C | 28. E |
| 4. A | 29. E |
| 5. A | 30. E |

- | | |
|-------|-------|
| 6. E | 31. E |
| 7. D | 32. E |
| 8. E | 33. E |
| 9. C | 34. C |
| 10. B | 35. E |
| 11. C | 36. C |
| 12. E | 37. B |
| 13. C | 38. E |
| 14. B | 39. A |
| 15. D | 40. A |
| 16. A | |
| 17. D | |
| 18. D | |
| 19. A | |
| 20. E | |
| 21. C | |
| 22. B | |
| 23. C | |
| 24. E | |
| 25. E | |